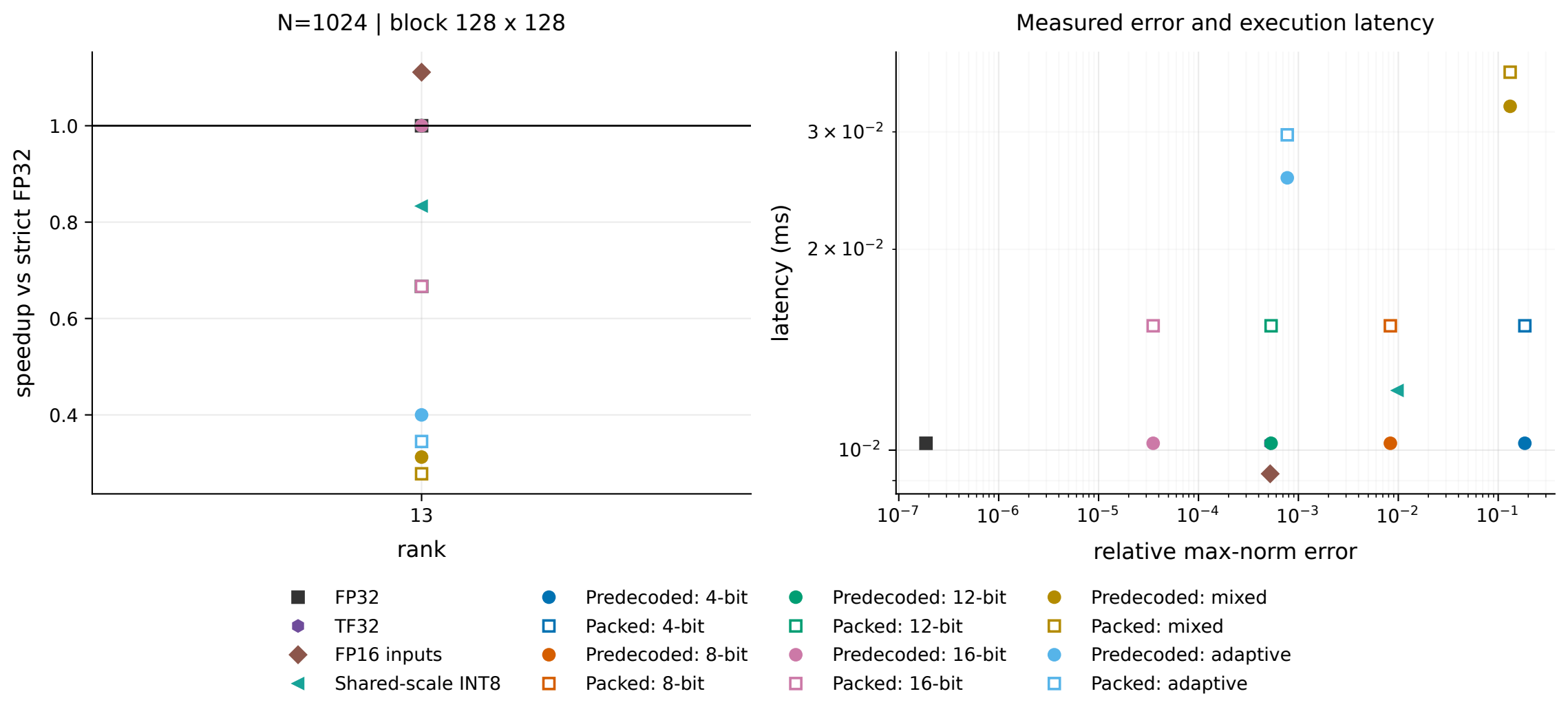
N=1024 | block 128 x 128
speedup vs strict FP32
1.0
0.8
0.6
0.4
13
rank
Measured error and execution latency
latency (ms)
$3 \times 10^{-2}$
$2 \times 10^{-2}$
$10^{-2}$
$10^{-7}$
$10^{-6}$
$10^{-5}$
$10^{-4}$
$10^{-3}$
$10^{-2}$
$10^{-1}$
relative max-norm error
FP32
TF32
FP16 inputs
Shared-scale INT8
Predecoded: 4-bit
Packed: 4-bit
Predecoded: 8-bit
Packed: 8-bit
Predecoded: 12-bit
Packed: 12-bit
Predecoded: 16-bit
Packed: 16-bit
Predecoded: mixed
Packed: mixed
Predecoded: adaptive
Packed: adaptive